

Engaging Users Through Play: Exploring Interactive Methods for Collecting Feedback in HCI

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Abstract

In Human-Computer Interaction (HCI), where conducting user studies is a key element, effectively engaging users to gather meaningful feedback remains a challenge. Methods usually used, such as surveys and questionnaires, often struggle with participation rates and lead to survey fatigue, limiting the quality of insights gathered. This paper explores three playful interactive methods for collecting user feedback in HCI as an alternative to these conventional approaches. We applied these methods in three workshops to assess their potential for improving participant engagement and the feedback collection process.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

Playful Methods, User Engagement, Feedback Collection

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1 Introduction

In recent years, traditional feedback collection surveys have faced scrutiny due to their lack of participation and potential to produce lower-quality data (cf. [5, 7, 17, 19]). Harms et al. [14] emphasize the importance of addressing this issue, noting that conventional survey designs are frequently seen as boring and unengaging, which can lead to a negative respondent behavior. For example, participants may rush through surveys without thoughtful responses or abandon them before completion (cf. [13]). Therefore, it is a challenge to motivate respondents to participate in surveys and to give their full attention while completing them, when they are perceived as boring (cf. [20]). These observations suggest that the incorporation

of playful and interactive elements into surveys can address some of the limitations of traditional feedback collection methods [20]. The evolution of digital technologies and interactive techniques has paved the way for more engaging and playful approaches to data collection.

In this paper, we present and explore three methods – *Emoji Feedback Method*, *Decision-Influenced Questionnaire*, and *Interactive Grid Questionnaire* – designed to maintain respondent engagement through playful and interactive elements. The *Emoji Feedback Method* leverages emojis combined with a voting mechanism to provide a playful way of giving feedback. The *Decision Influenced Questionnaire* method integrates principles of interactive storytelling into questionnaires to improve user engagement. The *Interactive Grid Questionnaire* method incorporates a puzzle-solving experience into the feedback process. We compare and explore these three methods to identify: *How does the use of emojis, interactive storytelling principles, and gamification influence the engagement of respondents in digital questionnaires?*

With this Late-Breaking Work, we discuss how these methods can enhance respondent engagement with the goal to offer a starting point for researchers and practitioners to further explore and refine these feedback collection techniques.

2 Related Work

This section presents studies addressing playful feedback methods and discusses broader topics such as gamification, participatory design, and user experience principles, emphasizing their relevance to playfulness and user engagement.

Feedback on Playful Systems and Activities. Existing approaches (e.g., [1, 8, 12, 24]) explore how participants provide feedback on playful systems, tools, or experiences. The focus is, however, on feedback about playfulness rather than making the feedback-giving process itself playful. For example, the study conducted by Alhusayen [1] investigated feedback provided by children aged 7–12 on a playful learning website using two methods: a Smileyometer by Read and MacFarlane [21], and sticky notes. The Smileyometer allowed children to select faces corresponding to a Likert scale (1=Angry to 6=Excited). The feedback on sticky notes was grouped into positive and negative categories based on keywords like *happy* and *boring*. Fagundo-Rivera et al. [8] collected feedback through a virtual survey after participants engaged in an educational escape room experience. The work by Fung and Chung [12] employed surveys to collect feedback on children’s playfulness. Schiavo et al. [24] – to mention a further example – used surveys to gather

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feedback on gamification topics but did not explore playful methods for feedback collection.

Gamification and Engagement Strategies. Several studies (e.g. [3, 15, 23]) examine gamification and interactive design strategies that enhance user engagement and motivation. While these works do not focus on feedback-giving, they provide insights into techniques that can inform the design of playful feedback methods. For example, Kuts [15], provides a comprehensive review of user interface characteristics that can be considered playful. Based on this review, the author proposes a model of user interface components designed to facilitate playful interactions, particularly useful for analysing playfulness in applications and services. While the review discusses scenarios where systems or applications deliver feedback in a playful manner to enhance user experience, it does not address the playful aspects of users engaging in feedback-giving activities. Rogers et al. [23] examine how to design engaging and intuitive systems, identifying key factors such as attention, pace, play, interactivity, conscious and unconscious control, engagement, and narrative style [23, p. 19]. Bødker [3] explores methods that address users' cultural, emotional, and experiential dimensions. This broader perspective emphasizes designing for creativity, enjoyment, and context-specific experiences alongside traditional functional goals. These insights align with the objectives of this work – the development of playful feedback methods that prioritize user engagement.

Feedback-Giving in Non-Playful Contexts. There are also studies (e.g., [16, 22]) where users provided feedback, but playfulness was integrated in other aspects of the study rather than in the feedback-giving process itself. For example, Liao et al. [16] included feedback provided to a chatbot. The playful element emerged through participants' off-task interactions with the chatbot, rather than through the feedback-giving process. Participants could provide feedback by typing “#fail” when the chatbot gave an incorrect answer. However, this study examined playful interactions with the system rather than playful methods for user-driven feedback-giving. Similarly, Rode et al. [22] collected feedback from children after navigating a telepresence robot, using printed questionnaires that included a Smileyometer (adapted from Read and MacFarlane [21]). However, the feedback-giving process was not designed to be playful.

While prior research has explored playful systems, engagement strategies, and feedback-giving methods, to our best knowledge, there remains a gap in investigating how playful feedback mechanisms can be designed and evaluated. This work aims to fill this gap by exploring methods that make the feedback-giving process itself more engaging and interactive.

3 Playful Interactive Methods

To address the lack of engagement that traditional surveys have, the following three playful interactive feedback methods were designed to explore their potential for respondent engagement.

3.1 Emoji Feedback Method

The *Emoji Feedback Method* (EFM) builds on the ability of emojis to visually convey emotions and simplify the feedback process. The EFM is designed to allow respondents to answer using only emojis

to encourage creative and expressive feedback. Participants respond to questions by typing answers into a response text box, combining emojis that best represent their thoughts and opinions. To increase interactivity, the method includes a voting phase after each question, in which respondents review and vote on other participants' anonymous answers that match their own views. This voting activity aims to introduce interactivity and collaboration, encourage engagement and get respondents to reflect on their own responses and those of others. The voting results highlight areas of shared insight or group consensus and provide a clearer understanding of the feedback gathered.

For our evaluation of this method, we selected Mentimeter [18] (see Figure 1a) because of its seamless integration of voting features that allow participants to upvote responses live during the session.

Challenges and Limitations. The interpretation of emojis can vary between individuals and devices, which may introduce ambiguity in the collected data. To minimize such misinterpretations, a possible solution is to ensure that all respondents have an identical version of emojis and use the same device, thereby mitigating the issue of respondents seeing different visual variants of the same emoji character (as can occur across operating systems like iOS and Android).

Additionally, emoji-based responses may not be suitable for collecting detailed or nuanced feedback. While they work well for expressing emotions and impressions (e.g., “How did you feel during today's session?”), they are less effective for complex or topic-specific questions (e.g., “Which aspect of the activity was most challenging for you?”).

3.2 Decision Influenced Questionnaire

The Decision-Influenced Questionnaire (DIQ) integrates interactive storytelling principles to create a dynamic, non-linear questionnaire experience. The DIQ offers a sequence of interconnected screens guided by respondents' answers, creating a dynamic, non-linear pathway through the questionnaire (see Figure 2). Each screen serves as a node in a decision tree (see Figure 1b), where the answer chosen by the respondent determines the next screen.

For the evaluation, we selected Typeform [25] and Fillout [9] to test the method. These tools were chosen due to their popularity and ease of use in creating dynamic questionnaires. Both Typeform and Fillout are known for their ability to implement conditional logic, essential for applying the DIQ method. Since both tools are similar, we wanted to explore if there are any differences and to identify possible limitations.

Challenges and Limitations. The DIQ method is particularly well-suited for surveys where the question order is flexible, and personalisation of questions does not negatively impact the collected data. However, a major limitation is that it relies on predefined answers, making it primarily suitable for closed-ended surveys. If open-ended questions are necessary, they can be inserted as additional questions in which the subsequent sequence is determined by the preceding question and not by the open-ended answer itself.

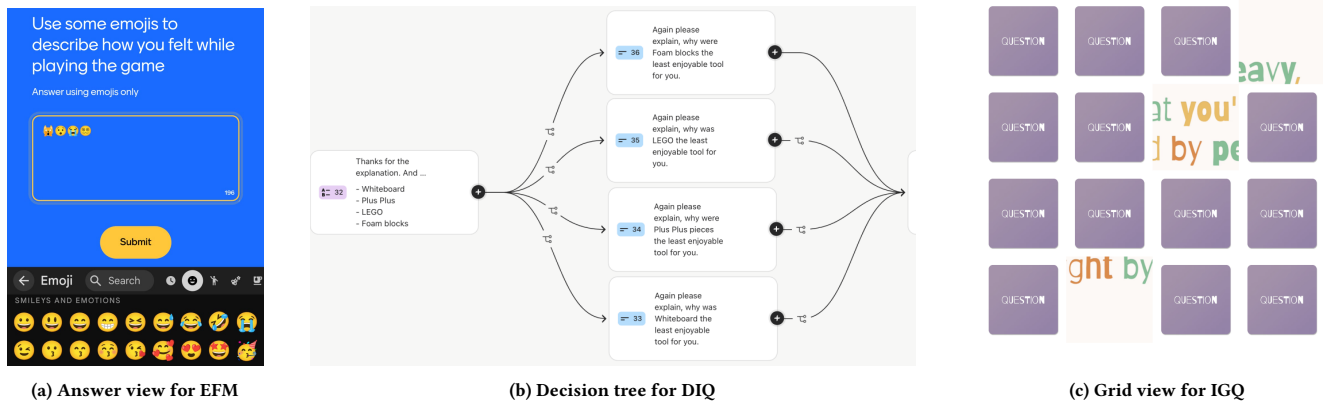


Figure 1: Examples which show a) answer view for EFM with the tool Mentimeter, b) a Typeform decision tree demonstrating how the survey dynamically adapts based on user responses, and c) web-based grid interface of IGQ where three questions were answered and the background image is partly uncovered.

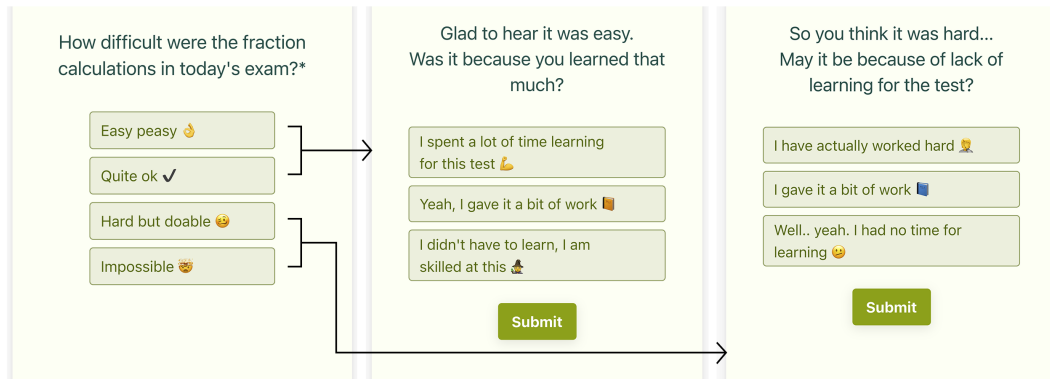


Figure 2: Two variants of DIQ flow, determined by a selected option.

3.3 Interactive Grid Questionnaire

The *Interactive Grid Questionnaire* (IGQ) embeds feedback questions into a grid-based interface, where each question is represented as a clickable tile on a web-based grid (see Figure 1c). A hidden background image is progressively revealed as respondents answer questions, adding a puzzle-solving element to the feedback process.

For the evaluation of this method, two tools are used. The first is Framer [11], a design and prototyping tool that enables the creation of interactive web-based interfaces. Framer was selected due to its capability to support complex user interactions, including the dynamic revealing of elements, which is central to the IGQ method. The second tool is Formspark [10], an online form handling service that captures and processes form submissions without requiring a dedicated back-end server.

Challenges and Limitations. As the order of the questions is determined by the respondents, the IGQ method is particularly suitable for surveys in which the order of the questions has no influence on the validity of the answers. For this reason, the designer must ensure that the questions can be answered independently of each

other. If a strict order of questions is required, the IGQ method could be further customized by triggering the questions in a predefined order, regardless of which tile is selected.

4 Evaluation

User studies were conducted to gather initial feedback on the three methods. As a use case, we integrated these methods into three workshops, originally designed to explore Augmented Reality (AR) games and the design of markers. This provided an opportunity to evaluate the methods in a real-world setting. The workshops took place at a grammar school in the Czech Republic, involving eight school children (average age: 13; 6 female, 2 male) from the same class.

4.1 Procedure

After obtaining approval from the Research Ethics Committee, the school leadership agreed to participate and facilitated the recruitment of children for the workshops. The workshop conductors provided the school with a detailed description of the workshops

and consent forms, which the parents of interested children had to complete and sign. The consent form thoroughly informed the parents about the data to be collected during the workshops and how it would be handled. On the first workshop, the school provided eight children with signed consent forms and a school psychologist who supervised the workshops.

EFM. The EFM user study took place during Workshop I. Each participant was provided with a smartphone to play an AR game and for the following feedback session. After they played the game, the participants were briefed on the EFM procedure by one of the workshop conductors. Afterwards, the questions were shown on their phones and they were asked to answer these questions by using the emojis. After all participants submitted their answer, they voted on anonymous answers submitted by other participants and the vote results were presented to them. To reduce ambiguity of the emoji answers, right after the EFM questionnaire session the respondents were asked to fill out a new questionnaire, which had the same questions, but required to use words instead of emojis. After two weeks, at Workshop II, participants were asked to reflect on their experience with using emojis for feedback compared to using words.

DIQ. To compare the DIQ method across two different tools, we conducted the evaluation using Typeform (Workshop II) and Fillout (Workshop III). Workshop II focused on feedback from creative activities, including developing game storylines and designing AR markers. After completing these tasks, participants were introduced to the Typeform-based DIQ questionnaire and asked to provide feedback via smartphones. The aim of Workshop III was to collect feedback on an AR game and design of characters for the game. We applied the DIQ method for the first part of the workshop – playing the AR game. The same procedure was followed, with participants answering DIQ questions (this time created by Fillout) through their smartphones once they finished the game.

IGQ. In Workshop III we also applied the IGQ method. While we explored the DIQ method for the first part of the workshop (collect feedback about the AR game), we used the IGQ method for the second part – to collect feedback after the participants designed characters for the game. The feedback collected through the IGQ focused on participants' satisfaction with the creative process and provided insights into their design choices. Feedback collection was conducted individually, using a laptop equipped with a remote eye-tracking device (SensoMotoric Instruments (SMI) RED 250, operating at 250 Hz). We decided to use eye-tracking for this study to explore not only participants' engagement but also strategies, and behavior (e.g., how they scan the questionnaire or where do they put their focus at first). However, because of the time limit of the workshop, it was only possible to collect eye-tracking data from five respondents. Before they saw the questions via the IGQ method, the method and the eye-tracking device were introduced and a calibration procedure was carried out for each respondent to ensure a proper operation of the eye-tracking device.

4.2 Results

For the analysis of the collected data from the EFM and DIQ sessions, we used the thematic analysis following Braun et al.'s [4] framework.

For EFM, a structured qualitative coding framework was developed to systematically assess the similarity between emoji-based and text-based responses, following template coding principles [2]. This framework defined four levels of similarity, ranging from no similarity between responses to direct textual translations of emojis. Each emoji-text response pair was manually reviewed and assigned a similarity level based on these predefined categories. Throughout the process, notable patterns and variations were documented alongside the quantitative similarity results, providing deeper insights into the relationship between emoji-based and text-based feedback. For DIQ, thematic analysis [4] was used to identify recurring patterns in participant responses and explore how the interactive questionnaire influenced engagement. The responses were first open-coded to capture emerging themes, which were then grouped into broader thematic categories reflecting engagement levels, user preferences, and perceived benefits or drawbacks of the method. This process provided insights into how DIQ facilitated engagement and where potential issues, such as survey fatigue, emerged in participant feedback. The eye-tracking data from the IGQ session were analyzed by using a velocity-based algorithm [6]. Gaze data were segmented to determine which areas of the interface attracted attention first, how participants navigated between questions, and whether visual patterns changed over time. Additionally, mouse-click data were recorded to examine selection strategies, identifying whether participants systematically answered questions (e.g., row-by-row selection) or followed a scattered, exploratory approach. The combined analysis of gaze patterns and interaction behavior helped to determine how the grid format affected engagement and whether participants' attention decreased over the course of the questionnaire.

EFM. Participants appeared highly engaged and often put considerable effort into formulating their emoji-based responses. We observed that the activity was marked by lighthearted discussions and playful atmosphere. When asked "How did you enjoy emojis compared to text?", the answers also reflected a generally positive impression: "A lot more fun than using text" (selected by 2 participants (P)), "Better than using text" (3P), "Same as using text" (1P) and "I'd rather use text" (2P). However, participants often struggled to express their thoughts using only emojis. This challenge was reflected in their verbal comments and their responses to questions about the effectiveness of emoji-based communication. For example one participant said that "what I exactly need isn't here" and "it's so complicated". The comparison of the text and emoji-based responses (in total we collected 48 response pairs) showed that 15 response pairs (31%) indicated no similarity between emoji and text versions, 9 response pairs (19%) showed minimal overlap, 13 response pairs (27%) were similar, and 11 response pairs (23%) were direct translations of each other.

The voting feature introduced a competitive element that increased engagement, but also led to strategic voting behavior. Some participants prioritized voting for themselves or their friends instead of selecting the most accurate or representative response. For example, one participant remarked: "Wait, dude, I'm going to win".

DIQ. Participants generally appreciated the simplicity of selecting predefined responses, as opposed to typing out answers. One participant noted: "I like how I just select the answer. It's bothering to

write my answer down, especially explaining why I think something". When asked "How did you enjoy this questionnaire compared to others?" responses reflected a preference for selection-based input: *"I like that there are options and that I don't have to write the answers by myself"*. However, we also observed that the method had difficulty maintaining engagement. Reasons were some repetitive questions (e.g., *"There are some questions asked twice, I have already answered this one."*) and the questionnaire's length. For example, 5 out of 8 participants agreed that the questionnaire was too long.

In terms of comparing the used tools, both tools collected and presented the respondents' data in the same format, which made it easy to compare the data and had no impact on the analysis process. Furthermore, we observed that both tools provided the same experience for the participants and there were no differences in their usage. Therefore, the choice between Typeform and Fillout primarily affects the questionnaire development process, e.g., the use of conditional hiding, and not the user experience, format or type of collected data.

IGQ. Participants demonstrated initial curiosity about the grid and surrounding interface elements, spending an average of 8.4 seconds observing the environment before selecting their first question tile. Furthermore, participants followed three different strategies when selecting question tiles: 1) two participants were systematically choosing tiles row by row, starting from the top-left corner. Both participants observed the uncovered parts of the background image after the first two selections but increasingly skipped this step as the session progressed, 2) two participants avoided systematic patterns, instead selecting tiles scattered across the grid; and 3) one participant started with a scattered selections but switched to a row-by-row strategy after answering three questions. However, engagement decreased over time. One participant frequently checked the system clock and browser elements, suggesting a loss of interest as the questionnaire progressed. Additionally, the quality of responses decreased toward the end of the session. Earlier questions received more detailed and thoughtful answers, whereas later questions were often answered hastily or with minimal effort.

5 Discussion

The findings suggest that playful interactive elements enhance initial engagement but also introduce challenges in sustaining focus and maintaining the richness of collected feedback. The EFM successfully engaged participants through its playful design and emoji-based interaction, fostering a lively and enjoyable atmosphere. The voting mechanism introduced an interactive group dynamic that maintained participant enthusiasm during the feedback activity. The DIQ utilized adaptive and personalized elements allowing participants to influence subsequent questions with their answers. Despite these features, the method achieved only moderate engagement, as many participants did not recognize or appreciate the interactive personalization. Additionally, the reverse questioning contributed to monotony further reducing engagement. The IGQ initially encouraged exploratory behavior with its game-like, grid-based format, as participants observed the grid and employed various strategies. However, revealing the hidden background image did not motivate the participants very much and their attention waned.

The results indicate that while playful interactive elements enhance engagement, they also introduce challenges in maintaining focus and ensuring feedback quality. The three methods – EFM, DIQ, and IGQ – were chosen for their potential to address survey fatigue and disengagement [14, 20], with each method representing a different approach to interactivity. EFM provides a visually expressive alternative to text-based feedback through emoji-based input in combination with a voting mechanism. DIQ employs interactive storytelling principles to create a more adaptive and personalized questionnaire experience. IGQ integrates game mechanics by allowing users to reveal a hidden image as they progress, fostering a more exploratory form of engagement. These methods were selected to cover different aspects of interactivity, ensuring a variety of ways to capture feedback.

Three key criteria were used to evaluate these methods: expressiveness and richness of feedback, sustained engagement and practical feasibility. EFM encouraged playful interaction but had limitations in depth of response, as emoji-based inputs were sometimes unable to capture complex thoughts. In addition, the voting feature introduced an element of competition, but strategic voting behavior affected the reliability of results. DIQ provided structured responses and ensured consistency, but led to fatigue due to the repetitive question structure. IGQ initially encouraged exploration, but engagement decreased over time as participants became less detailed in their responses as the session progressed.

Each method also presents specific benefits depending on the research context. EFM is particularly useful in capturing emotional responses and can be applied in group settings, as it contains a group voting activity. DIQ is well suited for surveys that require personalized response flow, making it valuable for decision-making studies, usability testing, or psychological research. IGQ is most applicable when exploratory engagement is desired, such as in participatory design studies. Researchers seeking to implement playful feedback collection should consider their specific study goals, the intended user group, and the required response format when selecting an approach.

5.1 Limitations

A primary limitation of this study is the small and specific sample size ($n=8$, school children), which limits the generalizability of the findings. Therefore, it would be necessary for future studies to focus on larger and more diverse participant groups to validate engagement trends and refine these methods for broader applicability. In addition to sample size constraints, each method presented unique challenges.

EFM. The voting mechanism introduced a social dynamic where participants strategically voted for themselves or their friends, which diminished the credibility of the results. While this added a certain playful factor, it detracted from the real purpose of voting, which was to identify shared perspectives.

The participants found it difficult to express complex ideas using only emojis. This was reflected in their verbal comments and feedback as well as in the similarity analysis of emoji and text responses. This was also the reason why they needed longer to respond with emojis compared to when they used words. This poses a challenge for the application of EFM in time-critical environments.

DIQ. The target age group (12–13 years old) may have influenced the results, as younger participants might experience repetition and closed-ended question formats differently than older respondent groups.

The repetitive nature of the reverse questioning technique, while beneficial for ensuring data consistency, detracted from engagement and contributed to questionnaire fatigue.

IGQ. The grid was displayed in a standard browser window, exposing distracting elements like browser tabs, the URL address, and the bottom navigation bar. These elements drew attention from participants.

The need to use a headrest for precise eye tracking may have affected participants' comfort and possibly altered their natural behavior during the session.

The design of the IGQ, which allowed participants to choose the order of the questions, led to differences in the detail of the answers. There were more thoughtful answers to the first questions, while the later questions were often answered hastily.

5.2 Recommendations and Future Work

EFM. The voting process should be replaced by a complementary activity, such as immediate text responses after emoji responses, to increase the relevance and reliability of the data. This change can help mitigate the distraction caused by competitive dynamics.

Using a mix of emojis and short text explanations could achieve a balance between the playful design of the method and the need for richer, more actionable data. This hybrid approach could solve participants' difficulties in expressing complex ideas solely through emojis.

Further and longer studies are necessary to investigate the feasibility of implementing EFM to assess how the time commitment and interactive elements, such as voting dynamics, affect participant engagement and the reliability of the data collected.

DIQ. The questions should be designed in such a way that the impression of repetition is avoided and at the same time the consistency of the answers can be checked, e.g., by rephrasing or spacing out similar questions in the questionnaire.

In future studies, the DIQ should also be analysed with older population groups to determine whether the problems observed in younger participants - such as questionnaire fatigue - also occur in other age groups.

The adaptive aspects of the questionnaire should be strengthened to make participants more aware of how their answers influence subsequent questions and thus increase engagement.

IGQ. The total number of questions should be reduced to prevent participant fatigue and ensure consistent engagement throughout the session. A shorter questionnaire can also improve the quality of feedback by avoiding rushed responses.

Similar to DIQ, future studies should include participants from different age groups and settings to assess whether the observed limitations are specific to younger respondents or apply more generally.

6 Conclusion

This Late-Breaking Work explored the potential of three playful interactive methods to increase engagement of feedback-giving process. The initial feedback showed that the EFM was successful in creating an interactive atmosphere, but often lacked depth of response and introduced distractions through its voting mechanism. The DIQ's adaptive and personalized features ensured consistent responses but led to monotony, particularly with reverse questions. The IGQ encouraged participants to interact freely with the interface, but showed decreasing engagement over time, with shorter responses in later stages.

Whilst these methods showed the potential to increase engagement, challenges such as repetitive elements, distractions and a loss of focus during longer tasks proved to be areas that require further improvements.

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References

- [1] Asma Alhussayen, Weaam Alrashed, and Evi Indriasari Mansor. 2015. Evaluating the User Experience of Playful Interactive Learning Interfaces with Children. *Proceedia Manufacturing* 3 (Jan. 2015), 2318–2324. <https://doi.org/10.1016/j.promfg.2015.07.378>
- [2] Erik Blair. 2015. A Reflexive Exploration of Two Qualitative Data Coding Techniques. *Journal of Methods and Measurement in the Social Sciences* 6, 1 (Jan. 2015), 14–29.
- [3] Susanne Bødker. 2006. When Second Wave HCI Meets Third Wave Challenges. In *Proceedings of the 4th Nordic Conference on Human-computer Interaction: Changing Roles*. ACM, Oslo Norway, 1–8. <https://doi.org/10.1145/1182475.1182476>
- [4] Virginia Braun and Victoria Clarke. 2006. Using Thematic Analysis in Psychology. *Qualitative Research in Psychology* 3, 2 (Jan. 2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [5] Rosaline de Koning, Abdullah Egiz, Jay Kotecha, Ana Catina Ciuculete, Setthasorn Zhi Yang Ooi, Nourou Dine Adeniran Bankole, Joshua Erhabor, George Higginbotham, Mehdi Khan, David Ulrich Dalle, Dawin Sichimba, Soham Bandyopadhyay, and Ulrick Sidney Kanmounye. 2021. Survey Fatigue During the COVID-19 Pandemic: An Analysis of Neurosurgery Survey Response Rates. *Frontiers in Surgery* 8 (Aug. 2021), 690680. <https://doi.org/10.3389/fsurg.2021.690680>
- [6] Andrew T. Duchowski. 2017. *Eye Tracking Methodology, Theory and Practice* (third edition ed.). Springer International Publishing.
- [7] Gunther Eysenbach and Jeremy Wyatt. 2002. Using the Internet for Surveys and Health Research. *Journal of medical Internet research* 4 (Nov. 2002), E13. <https://doi.org/10.2196/jmir.4.2.e13>
- [8] Javier Fagundo-Rivera, Rocío Romero-Castillo, Miguel Garrido-Bueno, and Pablo Fernández-León. 2024. Innovative Methodologies in University Teaching: Pilot Experience of an Escape Room in Nursing Students. *Education Sciences* 14, 10 (Oct. 2024), 1097. <https://doi.org/10.3390/educsci14101097>
- [9] Fillout. 2024. Fillout: Form builder for custom surveys and feedback. <https://fillout.com> Accessed: 2024-10-14.
- [10] FormSpark. 2024. FormSpark: Simple Form Submission API. <https://formspark.io/>. Accessed: 2024-10-10.
- [11] Framer. 2024. Framer: Prototyping Tool. <https://www.framer.com/>. Accessed: 2024-10-10.
- [12] Wing Kai Fung and Kevin Kien Hoa Chung. 2024. Playfulness and Kindergarten Children's Academic Skills: Executive Functions and Creative Thinking Processes as Mediators? *The Journal of Creative Behavior* 58, 3 (2024), 342–355. <https://doi.org/10.1002/jocb.654>
- [13] Theo Downes-Le Guin, Reg Baker, Joanne Mechling, and Erica Ruyle. 2012. Myths and Realities of Respondent Engagement in Online Surveys. *International Journal of Market Research* 54, 5 (Sept. 2012), 613–633. <https://doi.org/10.2501/IJMR-54-5-613-633>

- [14] Johannes Harms, Stefan Biegler, Christoph Wimmer, Karin Kappel, and Thomas Grechenig. 2015. Gamification of Online Surveys: Design Process, Case Study, and Evaluation. In *Human-Computer Interaction – INTERACT 2015*, Julio Abascal, Simone Barbosa, Mirko Fetter, Tom Gross, Philippe Palanque, and Marco Winckler (Eds.). Springer International Publishing, Cham, 219–236. https://doi.org/10.1007/978-3-319-22701-6_16
- [15] Ekaterina Kuts. 2009. Playful User Interfaces: Literature Review and Model for Analysis. In *Proceedings of DiGRA 2009 Conference: Breaking New Ground: Innovation in Games, Play, Practice and Theory*.
- [16] Q. Vera Liao, Muhammed Mas-ud Hussain, Praveen Chandar, Matthew Davis, Yasaman Khazaeni, Marco Patricio Crasso, Dakuo Wang, Michael Muller, N. Sadat Shami, and Werner Geyer. 2018. All Work and No Play?. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3173577>
- [17] Vikas Menon and Aparna Muraleedharan. 2020. Internet-Based Surveys: Relevance, Methodological Considerations and Troubleshooting Strategies. *General Psychiatry* 33 (July 2020), e100264. <https://doi.org/10.1136/gpsych-2020-100264>
- [18] Mentimeter. 2024. Mentimeter: Interactive Presentation Tool. <https://www.mentimeter.com>. Accessed: 2024-04-28.
- [19] Vikas O'Reilly-Shah. 2017. Factors Influencing Healthcare Provider Respondent Fatigue Answering a Globally Administered In-App Survey. *PeerJ* 5 (Sept. 2017), e3785. <https://doi.org/10.7717/peerj.3785>
- [20] Jonathan Puleston. 2011. Online Research–Game On!: A Look at How Gaming Techniques Can Transform Your Online Research. *Shifting the Boundaries of Research* (Jan. 2011), 20–50.
- [21] Janet C. Read and Stuart MacFarlane. 2006. Using the Fun Toolkit and Other Survey Methods to Gather Opinions in Child Computer Interaction. In *Proceedings of the 2006 Conference on Interaction Design and Children*. ACM, Tampere Finland, 81–88. <https://doi.org/10.1145/1139073.1139096>
- [22] Jennifer A Rode, Yifan Feng, Hanlin Zhang, Ria Rosman, Amanda S. Bastaman, John King, Madeline H. Samson, Xinyue Dong, Adam Walker, Matthew Horton, Janet C. Read, Martin Oliver, and Houda Elmimouni. 2024. Playful Telepresence Robots with School Children. In *Proceedings of the 13th Nordic Conference on Human-Computer Interaction (NordiCHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3679318.3685367>
- [23] Yvonne Rogers, Helen Sharp, and Jenny Preece. 2011. *Interaction Design: Beyond Human - Computer Interaction*. John Wiley & Sons.
- [24] Gianluca Schiavo, Eftychia Roumelioti, Giulia Deppieri, and Annapaola Marconi. 2024. Gamification Strategies for Child Protection: Best Practices for Applying Digital Gamification in Child Sexual Abuse Prevention. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference*. ACM, Delft Netherlands, 282–289. <https://doi.org/10.1145/3628516.3655813>
- [25] Typeform. 2024. Typeform: Beautifully designed forms and surveys. <https://www.typeform.com> Accessed: 2024-10-14.